

SYRINE SMATI

syrene.smati28@gmail.com | +216 51 054 359
linkedin.com/in/syrine-smati-a613702b4 | github.com/syrinesmati | syrinesmati-portfolio.vercel.app

EDUCATION

National Institute of Applied Science and Technology (INSAT)
Software Engineering Cycle (Master's Level)

Sept 2024 – Present
Expected Graduation: 2027

National Institute of Applied Science and Technology (INSAT)
Mathematics, Physics, and Computer Science Preparatory Cycle

Sept 2022 – June 2024

Les Pères Blancs High School

Baccalaureate in Mathematics, **Highest Honors** – Grade: 18.73/20

2022
Ranked 1st in Tunis 2 Region

PROFESSIONAL EXPERIENCE

TechTroll – AI Intern, New Cairo, Egypt

Jun 2026 – Present

- Built a **production document extraction pipeline** over **8 financial and regulatory document types** for **OSSUS.ai**, a sovereign AI GRC platform serving fintechs and regulated firms under Saudi financial regulation, with type-specific parsing/extraction strategies tailored to each document's structure.
- Developed **LLM-based compliance reasoning** components that score extracted evidence against **SAMA CSF** controls with pass/fail justification; benchmarked 8 LLMs across 36 test cases and selected the production model, raising prompt pass rate to **83%**.
- Building a policy-document mutation engine that converts review findings into automated regulatory-alignment edits via index-based targeting and structured JSON operation plans (ongoing).

Welyne – AI Intern, Ariana, Tunisia

Aug 2025 – Sep 2025

- Improved arm-length prediction accuracy to **79%** by benchmarking MLP vs. CNN architectures on biometric inputs (gender, height, weight); model deployed to production inside **WeeFizz**, Welyne's body-measurement engine that reconstructs full morphological profiles to drive size recommendation for fashion e-commerce and uniform suppliers.

Yonnov'IA – AI Intern, Remote, Marseille, France

Jun 2025 – Aug 2025

- Engineered a hybrid recommender system for Odoo eCommerce combining collaborative and content-based filtering, increasing recommendation coverage by **30%** across the product catalog through model tuning.
- Validated system performance on a catalog of **1,000+ products**; **deployed to production** and currently in use by Odoo clients.

PROJECTS

Tunisian Arabic Spoken Dialogue System – ASR & LLM

Academic Year 2025–2026

- Co-designed an end-to-end spoken dialogue system for Tunisian Arabic combining ASR, RAG retrieval, and a dialect-adapted LLM, shipped as an MVP with a React frontend and streaming FastAPI backends. Benchmarked 10 ASR models zero-shot and fine-tuned omniASR-LLM-3B on 344h of Tunisian speech, cutting WER by **53%** ($0.68 \rightarrow 0.38$). Selected Aya-Expanse-8B via a 5-model benchmark and adapted it through QLoRA continued pre-training (85M tokens) plus supervised fine-tuning (31.7K pairs), reaching **76%** token accuracy; released open-source as **TounsILM-8B** on Hugging Face.

Respiratory Sound Classification – Audio Spectrogram Transformer

Academic Year 2025–2026

- Reproduced and extended a published **AST + SAM** pipeline on the **ICBHI 2017** benchmark (6,898 annotated respiratory cycles, severe class imbalance), with mel-spectrogram preprocessing, cyclic padding and weighted resampling. Fine-tuned an **86M-parameter Audio Spectrogram Transformer** to **71% sensitivity** at 61% specificity, beating the reference paper's 68.3% at equal specificity. Ablated **Adaptive SAM**, Asymmetric Loss and Mixup, diagnosing an over-aggressive focusing exponent that traded sensitivity for specificity.

Tunisian Real Estate Price Prediction – Data Mining

Academic Year 2025–2026

- Built a price prediction pipeline over **25,646** listings aggregated from Kaggle and scraped from two Tunisian listing sites, with regex recovery of surface/rooms/type from free-text descriptions, Ward hierarchical clustering for price imputation, and 300+ region spellings harmonized into 239 canonical zones. Benchmarked **5 regression models** (Ridge, RandomForest, GradientBoosting, **XGBoost**, **LightGBM**) in a **Scikit-learn** pipeline with **MLflow** tracking; RandomForest reached $R^2 = 0.88$ on rentals (MAE 498 TND) and 0.73 on sales.

GitHub Trends Analyzer – Lambda Architecture for Big Data & Real-Time Analytics

Academic Year 2025–2026

- Designed a Lambda architecture unifying batch and speed layers over GitHub activity data. Batch layer: weekly Airflow DAG ingesting ~3–4M events/day from GH Archive via **PySpark**. Speed layer: Python producer polling the GitHub Events and Repos APIs every 10s (360 calls/hr) into **Kafka**. Merged both views into unified **HBase** tables powering a **Streamlit** dashboard; containerized with **Docker**.

SKILLS

Machine Learning and Deep Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, Hugging Face Transformers, OpenCV, YOLO

LLM Systems: prompt engineering, LLM benchmarking, LLM evaluation, retrieval-augmented generation (RAG), QLoRA fine-tuning

Big Data: Apache Spark, PySpark, Apache Airflow, HDFS, Apache HBase, Apache Kafka

MLOps: Docker, MLflow

Fullstack Development: React, Next.js, NestJS, FastAPI, .NET

Databases: PostgreSQL, MySQL, MongoDB, Redis, Neo4j

Programming Languages: Python, SQL, JavaScript, TypeScript, C#

Spoken Languages: English (Fluent), French (Native), Arabic (Native), Spanish (Beginner)